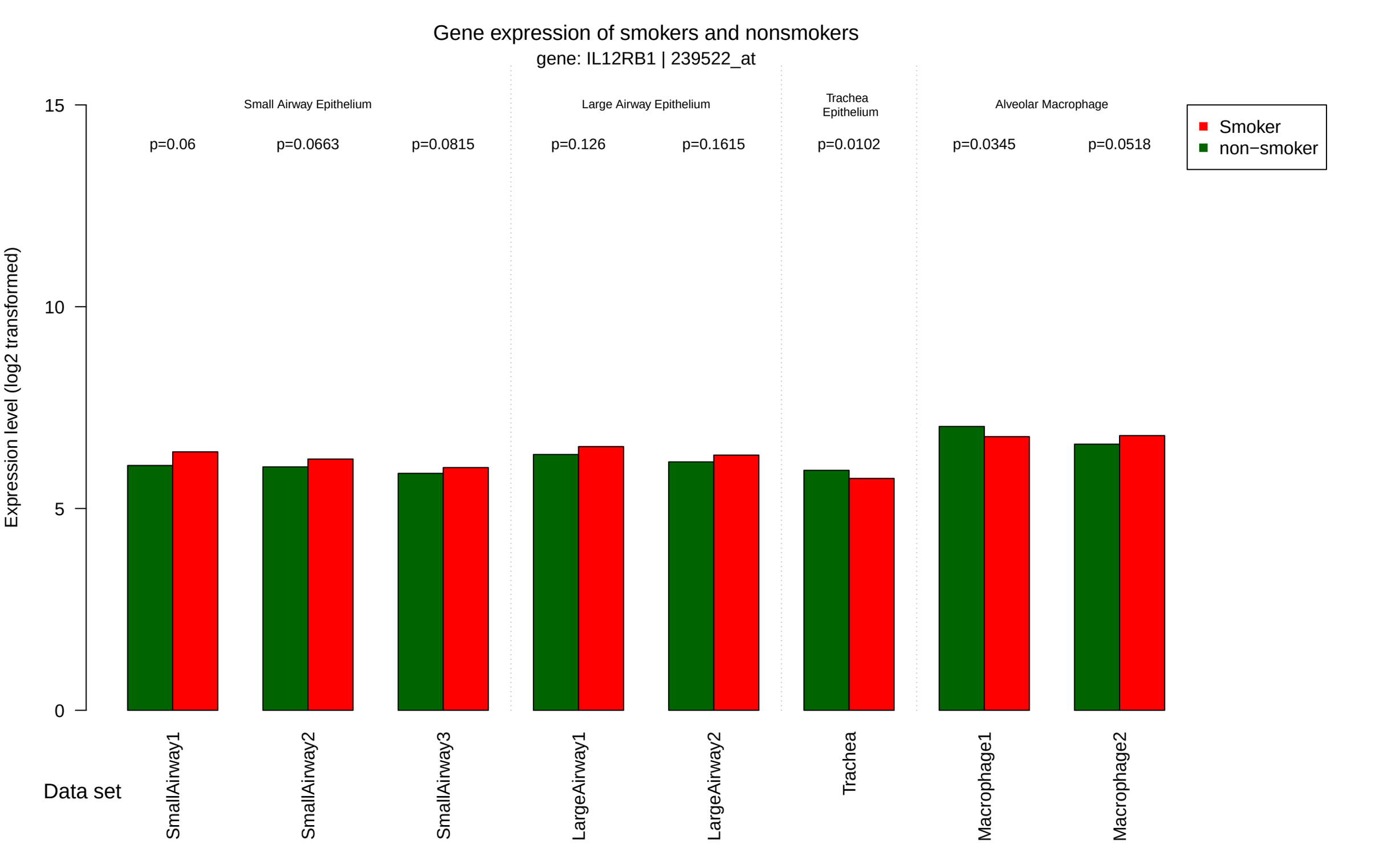
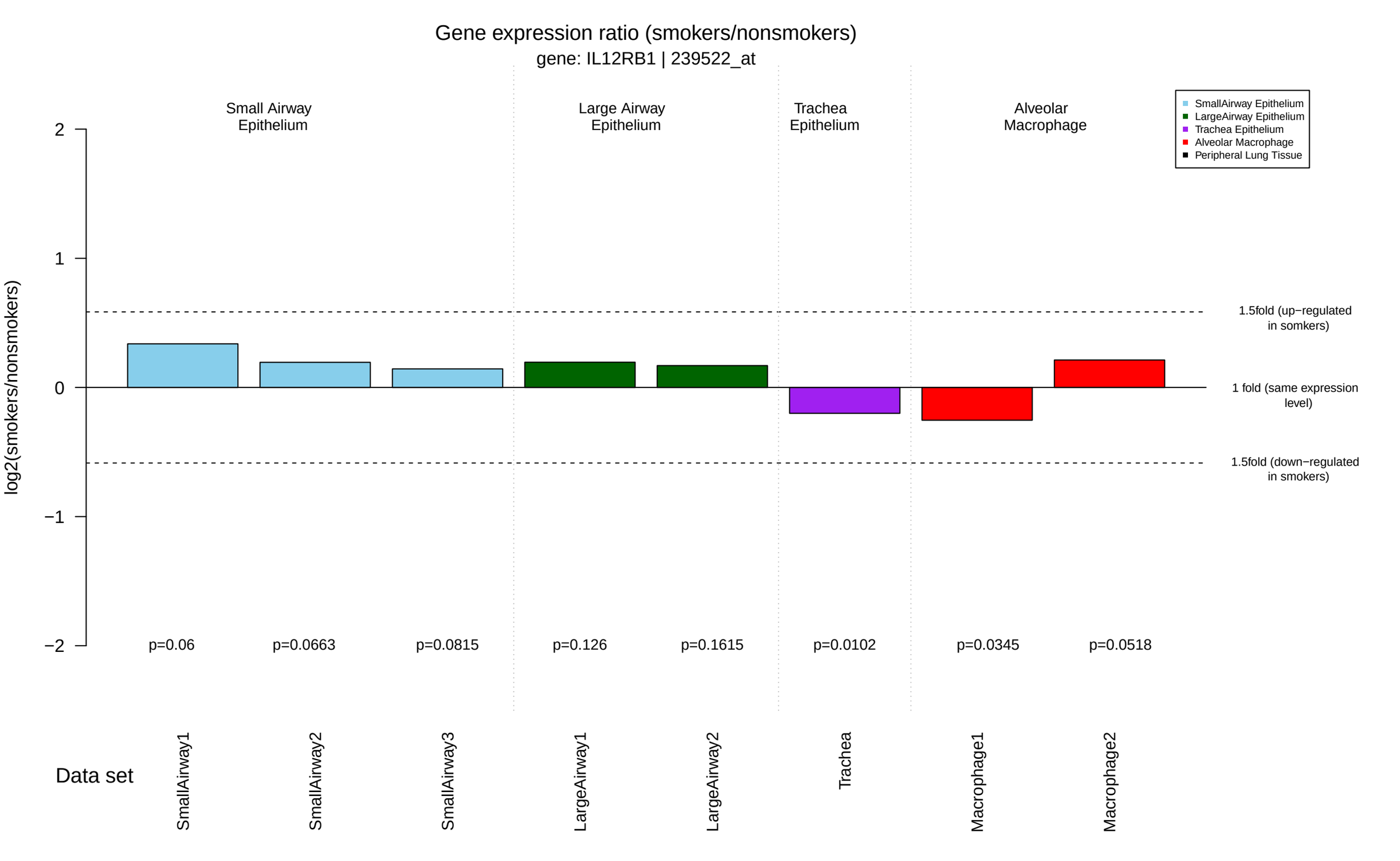
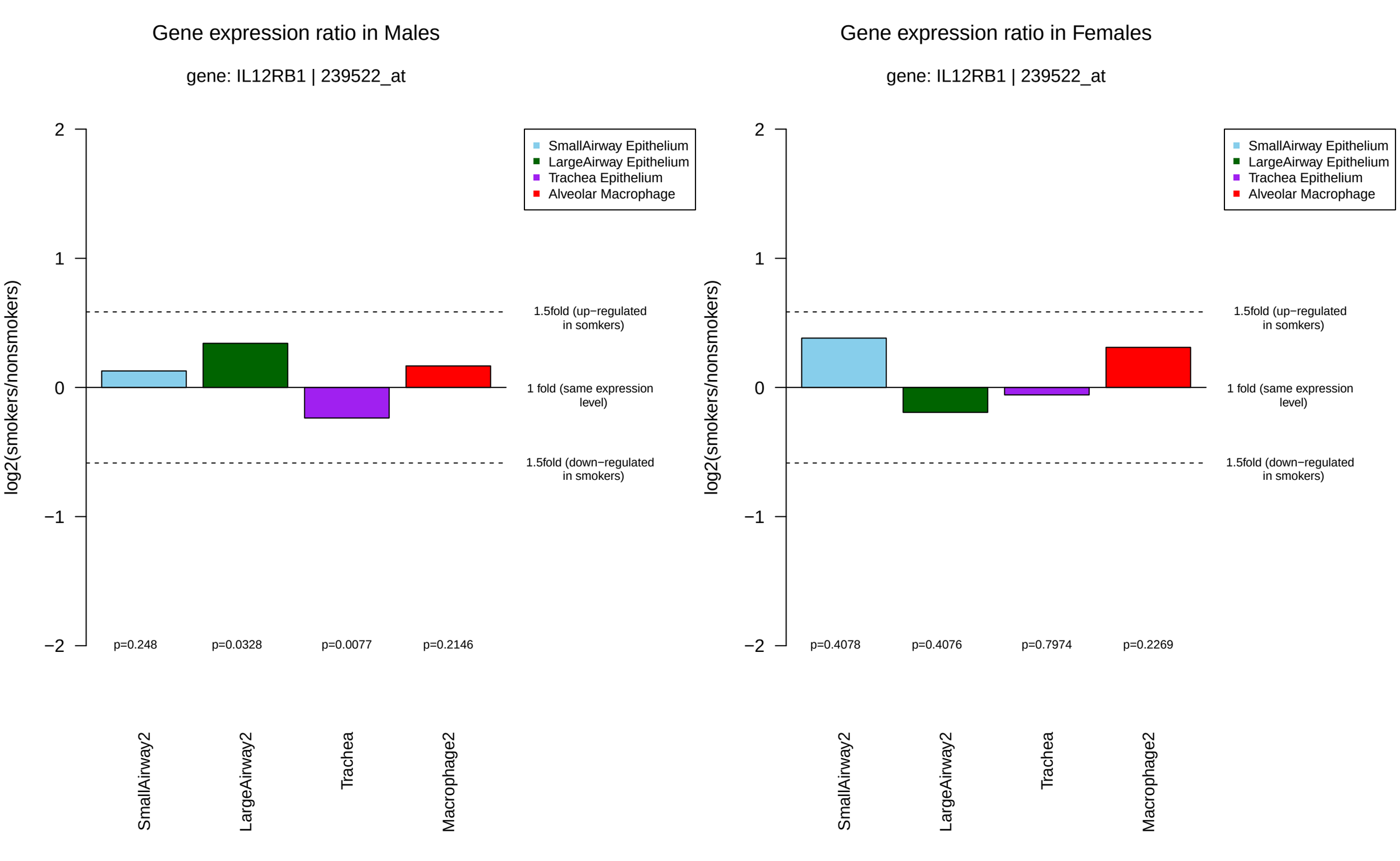
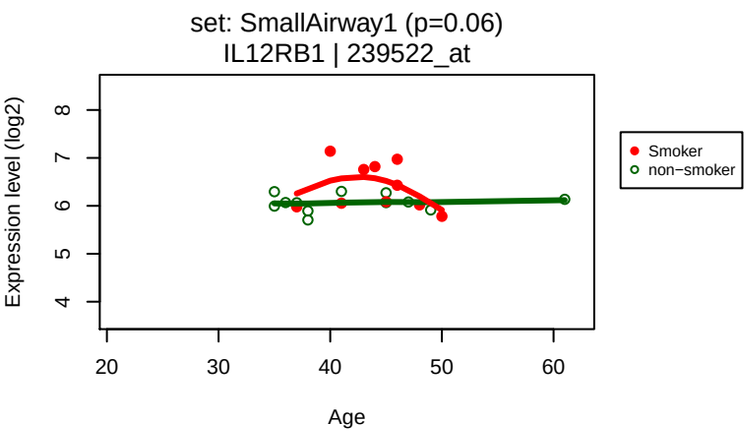
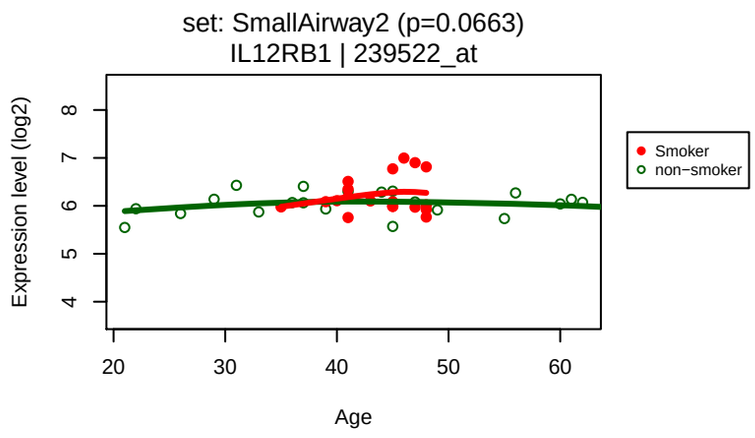
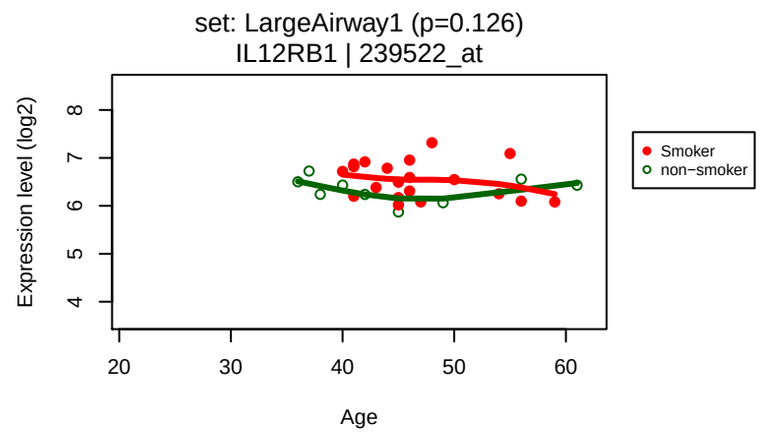

Gene expression of smokers and nonsmokers
gene: IL12RB1 | 239522_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.06
p=0.0663
p=0.0815
p=0.126
p=0.1615
p=0.0102
p=0.0345
p=0.0518
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

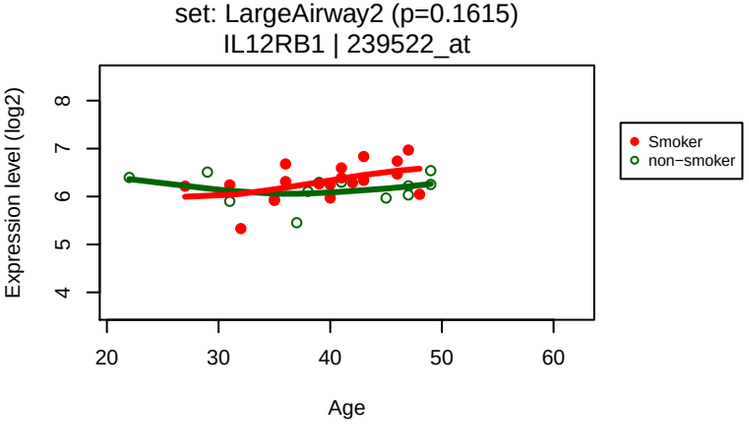

Gene expression ratio (smokers/nonsmokers)
gene: IL12RB1 | 239522_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.06
p=0.0663
p=0.0815
p=0.126
p=0.1615
p=0.0102
p=0.0345
p=0.0518
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

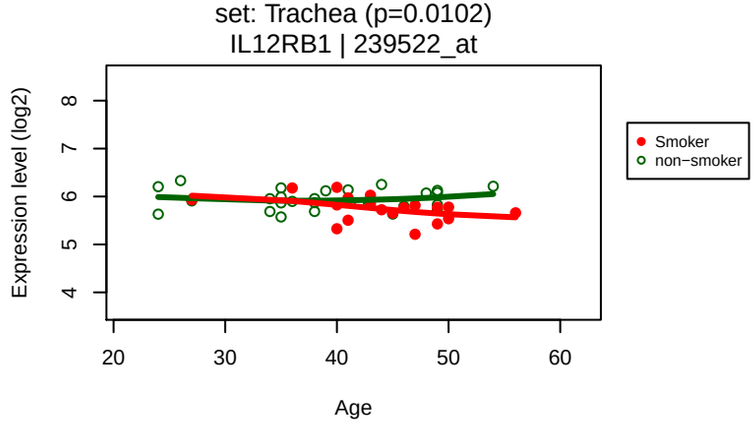

Gene expression ratio in Males
gene: IL12RB1 | 239522_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.248
p=0.0328
p=0.0077
p=0.2146
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL12RB1 | 239522_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4078
p=0.4076
p=0.7974
p=0.2269
SmallAirway2
LargeAirway2
Trachea
Macrophage2

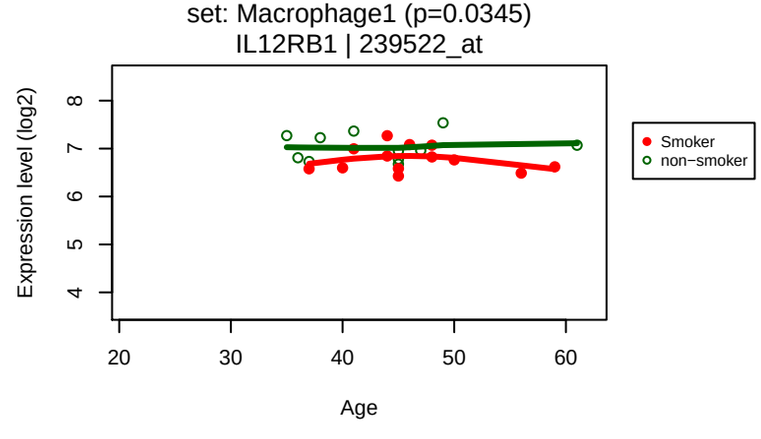

set: SmallAirway1 (p=0.06)
IL12RB1 | 239522_at
Expression level (log2)
Age
Smoker
non−smoker

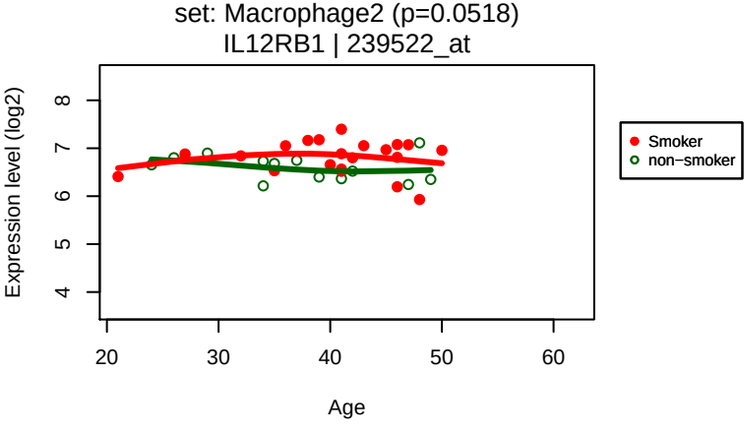

set: SmallAirway2 (p=0.0663)
IL12RB1 | 239522_at
Expression level (log2)
Age
Smoker
non−smoker

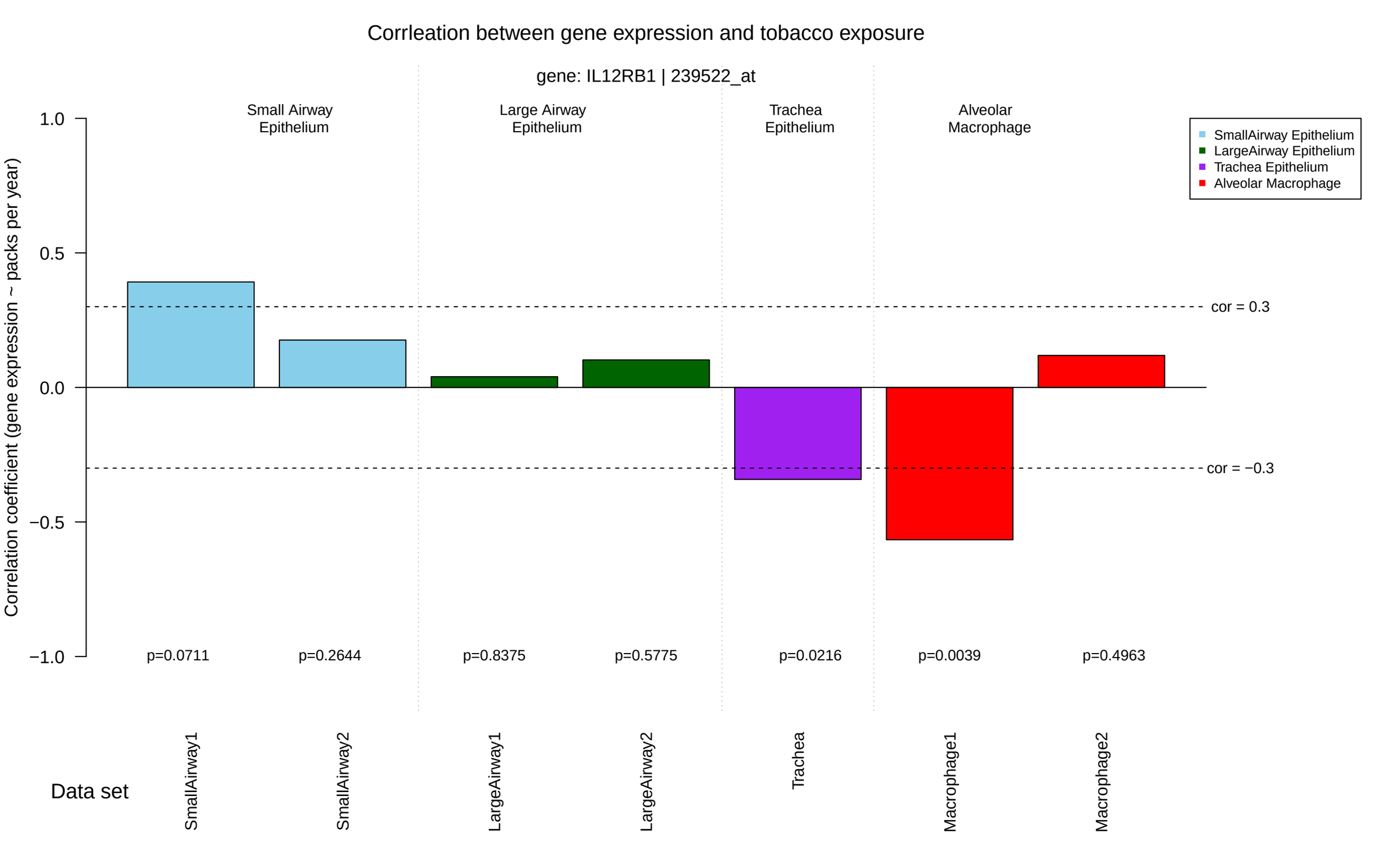

set: LargeAirway1 (p=0.126)
IL12RB1 | 239522_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1615)
IL12RB1 | 239522_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0102)
IL12RB1 | 239522_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0345)
IL12RB1 | 239522_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0518)
IL12RB1 | 239522_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL12RB1 | 239522_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0711
p=0.2644
p=0.8375
p=0.5775
p=0.0216
p=0.0039
p=0.4963
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2